EE 66 Fall 2026 — Signals, Dynamics, and Information — Discussion 11A

1. From LCCDE to State Space: A Second-Order Recursive Filter

Consider the causal discrete-time LTI system with input $x[n]$ and output $y[n]$ described by the linear constant-coefficient difference equation (LCCDE)

$$\forall n \in \mathbb{Z}, \quad y[n] = \frac{5}{6}y[n-1] - \frac{1}{6}y[n-2] + x[n].$$

Assume the system is initially at rest: $y[n] = 0$ for all $n < 0$.

(a) Draw a delay-adder-gain (DAG) block diagram that implements this LCCDE. Use only delay blocks (D), adders ($+$), and gain blocks. Label the outputs of your delay blocks.

(b) We now write this system in state-space form. Choose the outputs of the two delay blocks as the state variables:

$$\mathbf{q}[n] = \begin{bmatrix} q_1[n] \\ q_2[n] \end{bmatrix} = \begin{bmatrix} y[n-1] \\ y[n-2] \end{bmatrix}.$$

Write scalar equations for $q_1[n+1]$, $q_2[n+1]$, and $y[n]$ in terms of $q_1[n]$, $q_2[n]$, and $x[n]$.

(c) Express the system in matrix state-space form

$$\mathbf{q}[n+1] = A\mathbf{q}[n] + Bx[n], \qquad y[n] = C\mathbf{q}[n] + Dx[n].$$

Find A, B, C, D, and the initial state $\mathbf{q}[0]$.

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(d) The impulse response $h[n]$ is the output $y[n]$ when the input is $x[n] = \delta[n]$ and the system starts at rest. Use your state-space equations to compute $h[0]$, $h[1]$, $h[2]$, and $h[3]$.

(e) Now find a closed-form expression for $h[n]$.

i. Show that $h[0] = D$ and that $h[n] = CA^{n-1}B$ for $n \geq 1$.

ii. You are given that A has eigenvalues $\lambda_1 = \frac{1}{2}$ and $\lambda_2 = \frac{1}{3}$, with corresponding eigenvectors

$$\mathbf{v}_1 = \begin{bmatrix} \frac{1}{2} \\ 1 \end{bmatrix}, \qquad \mathbf{v}_2 = \begin{bmatrix} \frac{1}{3} \\ 1 \end{bmatrix}.$$

Diagonalize $A = V\Lambda V^{-1}$ and use this to find $h[n]$ for all n. Check your answer against the values from the previous part.

(f) Is this system BIBO stable? Justify your answer using the impulse response. How does your answer relate to the eigenvalues of A?